Zero-Shot Open-Vocabulary Scene Graph Pipelines on BIM-Derived RGB-D: A Controlled Study

BIM-Graphs Technical Report

BIM-Graphs Project

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Abstract

We evaluate whether open-vocabulary 3D scene graph pipelines can be applied zero-shot to RGB-D flythroughs rendered from Building Information Models. Using IFC as a source gives something no existing benchmark has: exact ground truth, since every pixel is traceable to a building element by construction. Our first result was that ConceptGraphs performs very poorly on such data. Our second, and more important, result is that this conclusion is not safe without a control: a matched run on Replica shows the same detector firing 25 objects per frame at confidence 0.49, against 1.93 at 0.26 on our renders, with 32% of our frames yielding no detection at all. The deficit decomposes into two independent causes. (i) Appearance: IFC supplies one flat colour per element, leaving 79% of each frame featureless against Replica's 21%; adding procedural surface texture reduces empty frames from 31.7% to 3.3% and nearly triples detections per frame. (ii) Content: Replica's top detections are pillow, sofa and coffee table, while ours are wall, door and window. A building model is largely unfurnished architecture, and object-centric pipelines discard walls and floors as background by construction — so they search for precisely what BIM omits and ignore what it states exactly. This is a structural mismatch that no rendering fidelity can repair. We quantify both causes with a three-condition appearance ladder, release the benchmark generator, and argue that room- and topology-level metrics, not object recall, are the meaningful measure for building-scale scene graphs.

1 Motivation

Open-vocabulary 3D scene graph methods are evaluated almost exclusively on ScanNet, Replica and Matterport, whose ground truth is hand-annotated and therefore imperfect and expensive. Evaluation consequently stops at object-level metrics, and connectivity — which room opens onto which, the property a robot actually navigates with — is essentially never scored, because no dataset states it reliably.

An IFC building model changes this. It provides exact geometry, exact element identity, and (after the topology recovery described in our companion literature review) exact room boundaries and connectivity. Rendering a flythrough from it therefore yields posed RGB-D with pixel-perfect instance masks and a fully known scene graph. That makes controlled ablation possible: pose noise, depth noise, appearance realism and furnishing density can each be varied independently, which cannot be done on a real scan.

This report asks the prerequisite question: do existing zero-shot pipelines work on such data at all, and if not, why not?

2 Benchmark construction

2.1 Ground truth generation

For each IFC model we build the hybrid hierarchical scene graph (rooms, elements, containment, door and passage connectivity), then plan a coverage trajectory that walks the recovered navigation graph. Two properties of the trajectory matter for fairness:

- It routes through doors using the recovered topology. Straight segments between room centres pass through walls, producing frames pressed against masonry that would penalise any method for reasons unrelated to perception.
- Frames whose median depth falls below 0.8 m are discarded, for the same reason.

Each frame stores RGB, uint16 depth in millimetres, a per-pixel instance mask keyed to the IFC GlobalId, the camera pose, and the room the camera occupied. Rendering is offscreen via EGL at 640×480 with a 58° vertical field of view, comparable to a RealSense D435.

2.2 Appearance conditions

Because we control the renderer, appearance realism becomes an experimental variable rather than a confound. We define three conditions:

A	flat	IFC diffuse colour only; one flat RGB per element.
B	textured	Procedural surface textures (plaster, wood, tile, carpet, glass, metal) selected by IFC class and material name, with UVs in world metres so pattern scale is physically consistent, plus oblique fill lights.
C	textured+clutter	As B, plus procedural furniture placed by room label. 98.5% of rooms in the corpus carry a functional label, so a kitchen can be furnished differently from a bedroom. Every placed item is registered as a ground-truth instance.

2.3 Reference dataset

As a control we use Replica room0 in the Nice-SLAM RGB-D rendering — the dataset ConceptGraphs itself was developed against. It is run through the identical detector, class list and configuration.

3 Experimental setup

Baselines. ConceptGraphs (object-centric open-vocabulary, YOLO-World + MobileSAM + CLIP ViT-H-14) and HOV-SG (hierarchical, with an explicit room layer). Both are the published implementations, not reimplementations.

Metrics. We report detector-level statistics (fraction of frames with no detection, detections per frame, mean confidence, distinct classes) and scene-graph-level statistics against exact ground truth.

Two decisions make the scene-graph metrics meaningful:

- Only observed ground truth is scored. An element the camera never saw cannot be recovered by any method; counting it as a miss would measure the trajectory. Visibility is read from the rendered instance masks.

- Objects and architecture are scored separately. ConceptGraphs discards wall/floor/ceiling as background by design, so scoring it against a ground-truth set dominated by walls measures a category mismatch, not method quality. We never merge the two buckets into one headline number.

We additionally report a hallucination rate: the fraction of predicted objects whose label is impossible in a building model (person, dog, …). This requires no matching and admits no ambiguity.

4 Results

4.1 The control: is it the method or the renderer?

Table 1 is the central result of this report.

Source	Frames	Empty frames	Det./frame	Confidence	Classes
A flat (model_0)	120	31.7%	1.93	0.26	27
B textured	60	3.3%	5.15	0.25	14
Replica (reference)	120	0.0%	25.02	0.49	60

Table 1: YOLO-World response under identical model, class list and thresholds. The detector never fails on Replica and fires 13× more often there than on flat BIM renders, at twice the confidence. A conclusion drawn from the flat condition alone would have been a statement about our renderer.

4.2 Why: low-level image statistics

Statistic	A flat	B textured	Replica
Gradient energy	0.0031	0.0103	0.0208
Local contrast	0.0177	0.0391	0.0763
Saturation	0.070	0.150	0.161
Distinct colours	35	91	306
Intensity entropy	3.59	3.96	5.40
Featureless area	79%	47%	21%

Table 2: Frame statistics (means over sampled frames). "Featureless area" is the fraction of 16×16 tiles with intensity standard deviation below 0.02. An IFC element carries a single diffuse colour, so a naive render is overwhelmingly flat — exactly the regime a detector trained on photographs has never seen. Texturing recovers roughly half the deficit in gradient and contrast, and matches Replica on saturation.

4.3 Scene-graph accuracy against exact ground truth

The two causes have distinct signatures. Varying appearance and content independently separates effects that a single "more realistic rendering" change would have confounded:

	Appearance (A→B)	Content (B→C)
Class accuracy	×2 on both models	—
Recall	unchanged (model_0)	+0.133 (model_0)
Precision	−0.08 on both	+0.21 (model_0)
Hallucination	barely moves	0.067 → 0.011

Texture fixes naming; content fixes finding. Neither substitutes for the other.

Cond.	Model	GT obj.	Pred.	Recall	Prec.	Class acc.	Halluc.
A flat	model_0	87	40	0.333	0.725	0.241	0.075
B textured	model_0	87	45	0.333	0.644	0.483	0.067
C +clutter	model_0	161	88	0.466	0.852	0.440	0.011
A flat	model_17	104	59	0.406	0.729	0.209	0.000
B textured	model_17	104	82	0.510	0.646	0.528	0.000
C +clutter	model_17	169	118	0.509	0.729	0.419	0.000

Table 3: ConceptGraphs against exact IFC ground truth, restricted to the object bucket and to instances actually observed by the camera. Condition C's ground truth is larger because injected furniture is itself registered as ground truth, so C-versus-B class accuracy is not like-for-like; recall, precision and hallucination are.

Texture changes naming, not finding. On model_0 the detector-level gain from texturing is large (1.93 to 5.15 detections per frame) yet the number of matched ground-truth objects is identical at 29. The additional detections are not new objects: they are more confident labels on regions already being detected, plus additional false positives, which is why precision falls. Read alone, the detector statistics would imply a recall improvement that did not occur.

Clutter helps only where content was missing. Adding furniture raises recall by 0.133 on model_0 but by 0.001 on model_17. This is what the content hypothesis predicts rather than a weakness of it: model_17 is an office whose rooms already contain furniture, so there was little deficit to correct, whereas model_0 is an open-corridor residential block whose coverage path spends much of its time on an unfurnished exterior deck. A uniform improvement across both buildings would have been the more suspicious result.

Hallucination tracks content, not appearance. The impossible-label rate on model_0 falls $0.075 \rightarrow 0.067 \rightarrow 0.011$. The person detections that survived filtering and merging in conditions A and B were not a quirk of the detector: they are what an object-seeking model does when aimed at empty architecture. Give it real objects and the phantoms largely stop.

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Building type dominates appearance. The gap between the two buildings under identical rendering (recall 0.510 vs 0.333 in condition B) exceeds the gap between appearance conditions within either building. A single corpus-wide average would therefore be misleading, and we report per-model figures throughout.

4.4 HOV-SG

HOV-SG is the more meaningful comparison because it models rooms explicitly. Its run is in progress at the time of writing: feature extraction over 51 keyframes completes in 4m43s, but sequential 3D mask merging requires approximately 165 s per iteration over 50 iterations (≈2 h15). We note this because an earlier attempt was terminated by our own 50-minute wall clock and could easily have been misread as a failure of the method.

One configuration detail is worth recording: create_graph.py skips hierarchical graph construction entirely for datasets named replica or scannet, on the assumption that those are

single-room scans. Because our renders adopt the Replica on-disk layout, registering them under that name would have produced a complete-looking run with the room layer — the only reason to use HOV-SG — silently absent.

4.5 The second cause: content, not appearance

Texture alone does not close the gap, and the class histograms show why:

Replica	pillow (690), sofa chair (249), coffee table (219), lamp (186), book (169), couch (147), shelf (142), potted plant (116)
BIM (B)	closet door (107), closet wall (67), window (63), picture (28), bed (14), person (11)

Replica rooms are furnished; IFC buildings are architecture. Across four storeys, model_0 contains 89 furniture instances in total, while a single Replica room presents hundreds of detectable objects. Since ConceptGraphs explicitly discards wall, floor and ceiling as background, it is searching for the category BIM does not model while ignoring the category BIM specifies exactly.

Hallucination. The label person appears in every BIM condition. There are no people in an IFC model. In the flat condition three person objects survived the three-detection filter and the merging stage, meaning the pipeline built stable 3D objects out of nothing. The rate falls with texture but does not vanish.

5 Discussion

The negative result was not safe. Our first measurement — six recovered objects for a four-storey building — looked like decisive failure. It was an artefact of frame stride: at 1.75 m spacing the median object was seen in exactly one frame, so multi-view association never fired. Dense sampling took the count from 6 to 40 and the median from 1 to 6 detections per object. A second apparent failure, near-total detector silence, proved to be flat rendering. Both would have been reported as method limitations.

What the field should take from this. Applying object-centric open-vocabulary pipelines to BIM is a category error, independent of rendering. The meaningful comparison is with methods that model rooms — HOV-SG and its successors — and the meaningful metric is room segmentation and connectivity, not object recall. Object recall on an unfurnished building measures the dataset.

Implication for our own pipeline. This supports scoping vision-language models to what IFC lacks — affordances, and functional labels for the 344 geometrically recovered rooms that carry no name — rather than to rediscovering objects that were never in the file.

6 Threats to validity

1. Synthetic appearance remains unmatched. Condition B closes about half the statistical gap to Replica, not all of it. Absolute scores on BIM renders should not be read as scores on real buildings.
2. Idealised sensing cuts the other way. Our poses are exact and depth is noiseless, which flatters the methods relative to a real robot. Appearance and sensing biases therefore act in opposite directions and must not be conflated.

3. Single-detector control. The Replica control isolates the detector, which is where the failure begins, but the association and fusion stages are not separately controlled.
4. Procedural clutter is not photoreal. Condition C tests the effect of content presence, not of realistic objects. It should be read as a lower bound on what furnishing contributes.
5. Scale. Results reported here cover a subset of the corpus; the generator runs over all 25 models.

7 Reproducibility notes

Both baselines are the published implementations. Getting them to run required resolving version drift that we record because it is likely to affect anyone repeating this work.

ConceptGraphs. Eight blockers: absent nvcc; CUDA 12.1 rejecting gcc>12; pip build isolation hiding torch from the chamferdist build; conda CUDA headers living under targets/x86_64-linux/include; supervision removing ColorPalette.default() (and being silently upgraded again by the GroundingDINO install); transformers too new for RAM; missing fairscale. Two defects were in the repository itself: a defaults: - base entry naming a file that does not exist, and a default match_method the code does not implement.

HOV-SG. numpy 2 breaking the compiled faiss; scikit-fmm requiring a C++ compiler; a hard-coded interactive matplotlib backend.

Modifications to third-party code. Four, all recorded in-line and none algorithmic: skip frames where YOLO-World returns no boxes (ultralytics SAM raises on an empty bbox tensor); skip the corresponding absent detection files during mapping; switch HOV-SG's matplotlib backend to Agg for headless operation; add a dataset branch so our Replica-layout data reaches HOV-SG's hierarchical graph construction, which is otherwise skipped for replica and scannet.

8 Conclusions

1. Zero-shot open-vocabulary pipelines do perform poorly on BIM-derived RGB-D, but the naive measurement overstates it by a wide margin. Controls are mandatory: without the Replica reference we would have attributed a rendering artefact to the method.
2. Appearance and content are separable causes with distinct signatures. Texture doubles class accuracy and leaves recall untouched; content raises recall and precision and reduces hallucination by a factor of six. Neither substitutes for the other, and a single "more realistic rendering" intervention would have confounded them.
3. Object-centric pipelines remain structurally mismatched to BIM: they discard walls and floors as background while seeking the furniture a building model does not contain. Injected clutter helps only where content was genuinely absent.
4. Building-scale scene graph evaluation should be room- and topology-centric. Exact IFC ground truth makes that measurable for the first time.

Errors caught during this study. We record these because each initially looked like a result. (i) A frame stride of five left the median object visible in one frame, suppressing multi-view association and yielding six objects for a four-storey building. (ii) Flat rendering silenced the detector on a third of frames. (iii) Detections per frame nearly tripled with texture while matched objects did not change at all. (iv) Our own evaluator scored correctly identified clutter

as class errors, making condition C appear to regress from 0.483 to 0.187 when the true figure is 0.440. (v) A 50-minute timeout killed HOV-SG mid-merge. Five apparent findings, all artefacts of setup or measurement.